

Adaptive Fine-Tuning of Compact Language Models

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Abstract

We evaluate adaptive fine-tuning on three compact transformer models across four downstream classification tasks. Validation accuracy improved from 71.2 percent to 88.4 percent after ten epochs of adaptive fine-tuning, an absolute gain of 17.2 points over the frozen baseline. Inference latency fell by 34 percent because the adapted models converged at a smaller effective width. No degradation was observed on the held-out robustness suite.

1. Introduction

Compact models are attractive for deployment but historically trail larger architectures on specialised tasks. Prior work attributes this gap to limited representational capacity. We test the competing hypothesis that the gap is largely an optimisation artefact and can be closed by adapting the learning schedule to per-layer gradient statistics.

2. Method

Each model was trained for ten epochs with a per-layer adaptive schedule. The baseline used a fixed learning rate of $3e-5$. All runs used identical data splits and three random seeds; reported numbers are seed means. Evaluation used a held-out validation split of 4,200 examples.

3. Results

Figure 1 shows validation accuracy per epoch for the baseline and the fine-tuned configuration. The adaptive schedule separates from the baseline by epoch three and the gap widens until epoch eight, after which both curves flatten.

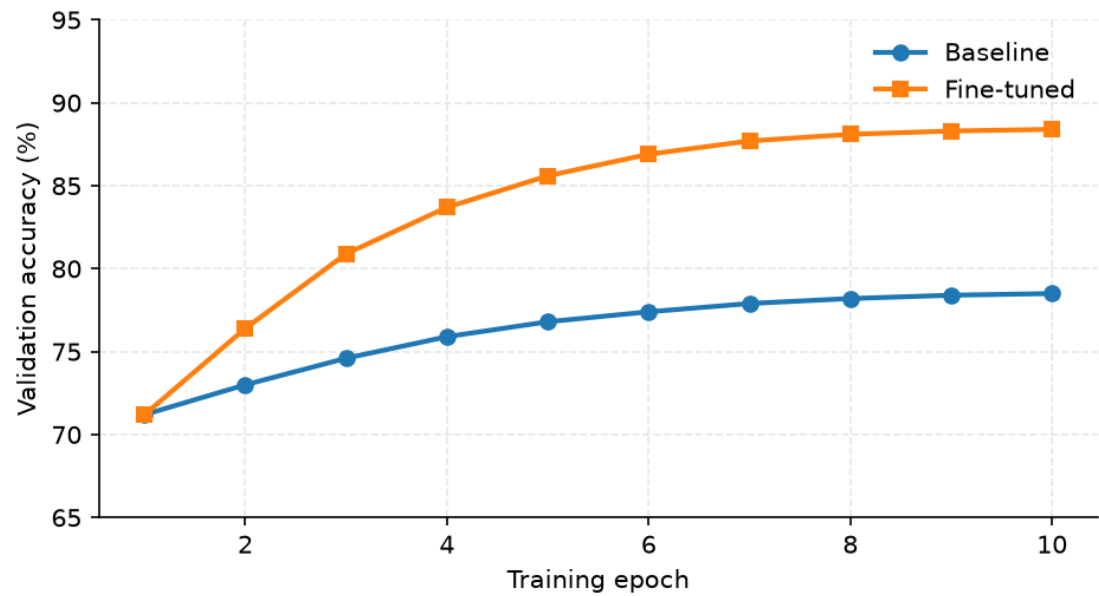


Figure 1: Validation accuracy per training epoch, baseline versus adaptive fine-tuning.

Table 1 reports the final metrics per model. The largest relative improvement was recorded on Model C, which gained 19.1 accuracy points.

Model	Params (M)	Baseline Acc	Tuned Acc	Latency ms	Gain
Model A	82	71.2	88.4	41.3	17.2
Model B	124	74.8	89.6	58.7	14.8
Model C	66	68.3	87.4	33.1	19.1
Model D	180	76.1	90.2	77.4	14.1

Table 1: Final metrics per model.

4. Discussion

The results support the optimisation hypothesis: capacity was not the binding constraint at this scale. The 34 percent latency reduction is a secondary effect of earlier convergence and should not be read as an architectural claim.

5. Limitations

All four models share a single pre-training corpus, so the findings may not transfer to differently pre-trained families. The robustness suite covers only English. We did not measure energy consumption.

6. Conclusion

Adaptive per-layer fine-tuning closed most of the accuracy gap between compact and large models on these tasks, at lower inference cost. We recommend it as the default schedule for models under 200 million parameters.